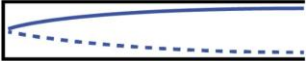


		$\frac{1}{4}\lambda$ $2r$ $\frac{1}{2}\lambda$
17	C	1/4 wave  1/2 wave  The fundamental wavelength will be shortened by half when changing from closed to open pipe. Since $v = f\lambda$, $f_{new} = 2f$.
18	A	Apply Principle of Superposition, the net vector summation of displacement at every point must be zero.